

ICT CSA Tutorial 02

- 01) Define the following memory terminologies: Bit, Nibble, Byte, and Word.
- 02) If a computer has 64 MB (megabytes) of byte-addressable memory, how many bits are needed in the memory address?
- 03) Explain the difference between Little Endian and Big Endian byte ordering conventions.
- 04) What are the two basic memory access operations required for executing a program, and briefly explain each?
- 05) A computer has a memory system with 216 locations, and each location stores 32 bits. What is the total memory capacity in Kilobytes (KB)?
- 06) If a computer has 2 GB of memory, and it is byte-addressable, how many bits are required for the memory address?
- 07) A 32-bit number is given as 11100000 01010101 10011001 00001111. If this number is stored in memory starting from address 3000 using **Little Endian** byte ordering, what will be the content at memory addresses 3000 and 3003?
- 08) Consider an instruction `ADD R1, LOCA` where LOCA is memory address 4000, the initial value of register R1 is 150, and the content of memory location 4000 (`Mem[4000]`) is 75. What will be the final value of R1 after the execution of this instruction?
- 09) An ARM embed-C cross-compiler running on a desktop PC generates ARM machine code. If the desktop PC uses a 64-bit architecture and the ARM embedded development board uses a 32-bit architecture, how many bits will be in the word generated by the cross-compiler for the embedded board?
- 10) List and briefly describe the three levels of computer languages discussed in the lecture.
- 11) Explain the functions of an Assembler and a Compiler.
- 12) What are the two special-purpose registers used for interfacing with primary memory, and what is the function of each?
- 13) What are the two special-purpose registers used for keeping track of programs/instructions, and what is their role?
- 14) Describe the basic steps involved in reading data from memory.
- 15) What is a "bus" in computer architecture, and how does single-bus architecture differ from multi-bus architecture in terms of data transfer?